



Machine Learning

Quiz-2 Solutions



Set 1: (Total marks 15)

B, C, B, D, B, C, C, B, B, D

The Quadrant-Shifted XOR problem uses the following data points:

Class (y)	Data Points ($\mathbf{x}$)
$y = +1$	$\mathbf{x}_1 = (2, -2), \mathbf{x}_2 = (-2, 2)$
$y = -1$	$\mathbf{x}_3 = (-2, -2), \mathbf{x}_4 = (2, 2)$

The kernel is the **homogeneous degree-2 polynomial kernel**: $K(\mathbf{x}, \mathbf{z}) = (\mathbf{x}^T \mathbf{z})^2$
 Feature Mapping $\Phi(\mathbf{x})$ For $\mathbf{x} = (x_1, x_2)^T$ and $\mathbf{z} = (z_1, z_2)^T$:

$$\begin{aligned} K(\mathbf{x}, \mathbf{z}) &= (x_1 z_1 + x_2 z_2)^2 \\ &= x_1^2 z_1^2 + 2x_1 x_2 z_1 z_2 + x_2^2 z_2^2 \\ &= (x_1^2)(z_1^2) + (\sqrt{2}x_1 x_2)(\sqrt{2}z_1 z_2) + (x_2^2)(z_2^2) \end{aligned}$$

The explicit feature map $\Phi(\mathbf{x}) : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is:

$$\mathbf{z} = \Phi(\mathbf{x}) = (x_1^2, \sqrt{2}x_1 x_2, x_2^2)^T$$

Mapping the four data points to the 3D feature space $\mathcal{H}$:

1. $\Phi(\mathbf{x}_1) = \Phi(2, -2) = (4, -4\sqrt{2}, 4)^T = \mathbf{v}_+$
2. $\Phi(\mathbf{x}_2) = \Phi(-2, 2) = (4, -4\sqrt{2}, 4)^T = \mathbf{v}_+$
3. $\Phi(\mathbf{x}_3) = \Phi(-2, -2) = (4, 4\sqrt{2}, 4)^T = \mathbf{v}_-$
4. $\Phi(\mathbf{x}_4) = \Phi(2, 2) = (4, 4\sqrt{2}, 4)^T = \mathbf{v}_-$

The problem reduces to separating the two distinct points $\mathbf{v}_+$ (Class +1) and $\mathbf{v}_-$ (Class -1). Since they are distinct, they are **linearly separable** in the feature space $\mathcal{H}$. The kernel matrix $K_{ij} = (\mathbf{x}_i^T \mathbf{x}_j)^2$ is:

$$\mathbf{K} = \begin{pmatrix} (\mathbf{x}_1^T \mathbf{x}_1)^2 & (\mathbf{x}_1^T \mathbf{x}_2)^2 & (\mathbf{x}_1^T \mathbf{x}_3)^2 & (\mathbf{x}_1^T \mathbf{x}_4)^2 \\ (\mathbf{x}_2^T \mathbf{x}_1)^2 & (\mathbf{x}_2^T \mathbf{x}_2)^2 & (\mathbf{x}_2^T \mathbf{x}_3)^2 & (\mathbf{x}_2^T \mathbf{x}_4)^2 \\ (\mathbf{x}_3^T \mathbf{x}_1)^2 & (\mathbf{x}_3^T \mathbf{x}_2)^2 & (\mathbf{x}_3^T \mathbf{x}_3)^2 & (\mathbf{x}_3^T \mathbf{x}_4)^2 \\ (\mathbf{x}_4^T \mathbf{x}_1)^2 & (\mathbf{x}_4^T \mathbf{x}_2)^2 & (\mathbf{x}_4^T \mathbf{x}_3)^2 & (\mathbf{x}_4^T \mathbf{x}_4)^2 \end{pmatrix} = \begin{pmatrix} (8)^2 & (-8)^2 & (0)^2 & (0)^2 \\ (-8)^2 & (8)^2 & (0)^2 & (0)^2 \\ (0)^2 & (0)^2 & (8)^2 & (-8)^2 \\ (0)^2 & (0)^2 & (-8)^2 & (8)^2 \end{pmatrix}$$

The **full kernel matrix** is:

$$\mathbf{K} = \begin{pmatrix} 64 & 64 & 0 & 0 \\ 64 & 64 & 0 & 0 \\ 0 & 0 & 64 & 64 \\ 0 & 0 & 64 & 64 \end{pmatrix}$$

The dual problem is $\max_{\alpha} W(\alpha) = \sum \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K_{ij}$, subject to $\sum \alpha_i y_i = 0$ and $\alpha_i \geq 0$. Given the data structure ($\mathbf{x}_1 = \mathbf{x}_2$ and $\mathbf{x}_3 = \mathbf{x}_4$ in $\mathcal{H}$), we impose $\alpha_1 = \alpha_2$ and $\alpha_3 = \alpha_4$. The constraint $\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4 = 0$ implies $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \alpha$. The objective function simplifies to $W(\alpha) = 4\alpha - 256\alpha^2$. Setting $\frac{dW}{d\alpha} = 4 - 512\alpha = 0$ yields the optimal value:

$$\alpha^* = \left(\frac{1}{128}, \frac{1}{128}, \frac{1}{128}, \frac{1}{128} \right)^T$$

All four points are Support Vectors.

The decision function is $f(\mathbf{x}) = \sum_{i=1}^4 \alpha_i^* y_i K(\mathbf{x}_i, \mathbf{x}) + b$.

The weight vector is $\mathbf{w} = \sum_{i=1}^4 \alpha_i^* y_i \Phi(\mathbf{x}_i) = \frac{1}{128} [(+1)\Phi(\mathbf{x}_1) + (+1)\Phi(\mathbf{x}_2) + (-1)\Phi(\mathbf{x}_3) + (-1)\Phi(\mathbf{x}_4)] = \frac{1}{64} (0, -8\sqrt{2}, 0)^T = (0, -\frac{\sqrt{2}}{8}, 0)^T$.

Bias b

Using the KKT condition $y_1(\mathbf{w}^T \Phi(\mathbf{x}_1) + b) = 1$:

$$1 \cdot \left[\left(0, -\frac{\sqrt{2}}{8}, 0 \right)^T \cdot (4, -4\sqrt{2}, 4) + b \right] = 1$$
$$\frac{8}{8} + b = 1 \implies 1 + b = 1 \implies b = 0$$

The separating hyperplane is $\mathbf{w}^T \Phi(\mathbf{x}) + b = 0$:

$$\left(0, -\frac{\sqrt{2}}{8}, 0 \right)^T \cdot (x_1^2, \sqrt{2}x_1x_2, x_2^2)^T = 0$$
$$-\frac{\sqrt{2}}{8}(\sqrt{2}x_1x_2) = 0 \implies -\frac{1}{4}x_1x_2 = 0 \implies \mathbf{x}_1\mathbf{x}_2 = 0$$

Hence, the decision boundary is the union of the x_1 -axis and the x_2 -axis in the original space.

Scaling: Let $\mathbf{x}' = c\mathbf{x}$. The feature map becomes: $\Phi(\mathbf{x}') = c^2\Phi(\mathbf{x})$ The feature vectors are simply scaled by a positive constant c^2 . Since the relative positions (and hence separability) are maintained, **scaling does not affect linear separability** in the feature space $\mathcal{H}$ if $b = 0$.

Translation Let $\mathbf{x}' = \mathbf{x} + \mathbf{a}$. The kernel remains $K(\mathbf{x}', \mathbf{z}') = (\mathbf{x}'^T \mathbf{z}')^2$.

$$\Phi(\mathbf{x}') = ((x_1 + a_1)^2, \sqrt{2}(x_1 + a_1)(x_2 + a_2), (x_2 + a_2)^2)^T$$

Since $\Phi(\mathbf{x}')$ is not a simple translation $\Phi(\mathbf{x})$, the **homogeneous degree-2 kernel is NOT translation invariant**. Linear separability in $\mathcal{H}$ is generally **not guaranteed** for translated data using this specific kernel. The translated data this must be verified on a case-by-case basis.

Set 2: (Total marks 15)
C, C, B, B, D, B, C, B, D, B

Feature map for the homogeneous degree-2 polynomial kernel is same as Set-1. Mapped feature vectors are:

$$\begin{aligned}\phi(1,1) &= \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix}, \phi(-1,1) = \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix}, \phi(1,-1) = \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix}, \phi(-1,-1) = \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix}, \\ \phi(0,0) &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \phi(0.4,0.4) = \begin{bmatrix} 0.16 \\ 0.16\sqrt{2} \\ 0.16 \end{bmatrix}.\end{aligned}$$

The four +1 points map to only two distinct vectors. Because the positives are far from the origin and the negatives lie near the origin, linear separation in $\mathbb{R}^3$ is plausible. The full 6×6 kernel matrix is (calculations as in Set-1)

$$K = \begin{bmatrix} 4 & 0 & 0 & 4 & 0 & 0.64 \\ 0 & 4 & 4 & 0 & 0 & 0 \\ 0 & 4 & 4 & 0 & 0 & 0 \\ 4 & 0 & 0 & 4 & 0 & 0.64 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0.64 & 0 & 0 & 0.64 & 0 & 0.1024 \end{bmatrix}.$$

To solve the hard margin SVM dual form we use symmetry to reduce unknowns:

- Let $\alpha_1 = \alpha_4 = p$ (these correspond to $\phi(1,1)$ and $\phi(-1,-1)$).
- Let $\alpha_2 = \alpha_3 = q$
- Let $\alpha_5 = 0$ (the origin will not be a support vector - it is interior to the negative side).
- Let $\alpha_6 = r$ (for the $(0.4,0.4)$ point).

The equality constraint $\sum_i \alpha_i y_i = 0$ (labels +1 for $i = 1, \dots, 4$, -1 for $i = 5, 6$) reduces to

$$2p + 2q - r = 0 \implies r = 2(p + q).$$

Because of the $\pm\sqrt{2}$ symmetry in the second coordinate, optimality suggest $p = q$ or $r = 4p$. Compute w from the dual/primal relationship $w = \sum_i \alpha_i y_i \phi(x_i)$, but this problem is complex to solve analytically (not impossible!). The support vectors will be $(0.4,0.4)$ and one or two of the +1 points. A quick inspection of mapped features suggest

$$w \propto \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}; \quad b \neq 0.$$

So w points along $[1, 0, 1]$ in feature space, i.e. $\propto x_1^2 + x_2^2$ in input coordinates. Visually imagine -ive points and +ive points are separated by a circle. Note the number of data points in training set and hence the bias will impact the circle radius.